

8.8 Volumes with Cross Sections: Triangles and Semicircles

Calculus

Solutions

Practice

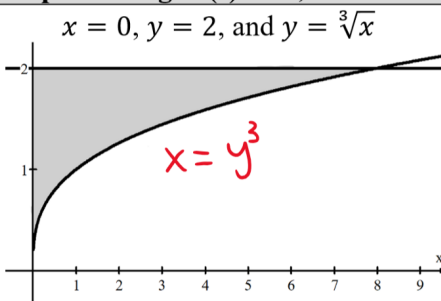
The bounded region shown for each problem represents the base of a solid. Find the volume of each solid based on the given cross sections. Set up the integral(s) first, then use a calculator to evaluate.

1. Semicircle cross sections perpendicular to the x -axis.

$$\int_0^8 \frac{\pi}{2} \left(\frac{2 - \sqrt[3]{x}}{2} \right)^2 dx$$

$$\frac{\pi}{8} \int_0^8 (2 - \sqrt[3]{x})^2 dx$$

1.2566



2. Equilateral cross sections perpendicular to the y -axis.

$$\int_0^2 \frac{\sqrt{3}}{4} (y^3)^2 dy$$

$$\frac{\sqrt{3}}{4} \int_0^2 y^6 dy \approx$$

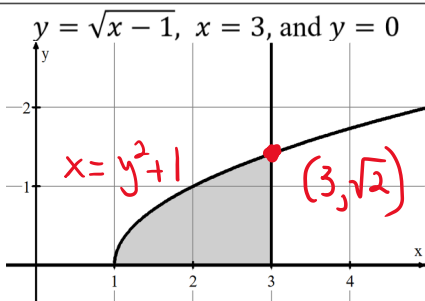
7.9179

3. Isosceles right triangle cross sections perpendicular to the x -axis.

$$\int_1^3 \frac{1}{2} (\sqrt{x-1})^2 dx$$

$$\frac{1}{2} \int_1^3 (x-1) dx$$

1



4. Semicircle cross sections perpendicular to the y -axis.

$$\int_0^{\sqrt{2}} \frac{\pi}{2} \left(\frac{3 - (y^2+1)}{2} \right)^2 dy$$

$$\frac{\pi}{8} \int_0^{\sqrt{2}} (3 - (y^2+1))^2 dy$$

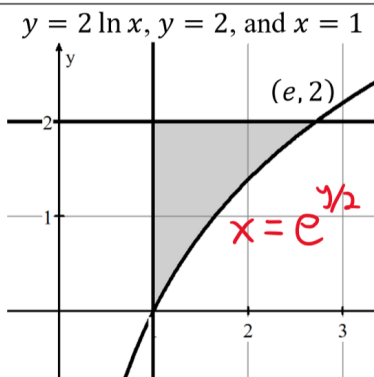
1.1847

5. Semicircle cross sections perpendicular to the x -axis.

$$\int_1^e \frac{\pi}{2} \left(\frac{2 - 2\ln x}{2} \right)^2 dx$$

$$\frac{\pi}{2} \int_1^e (1 - \ln x)^2 dx$$

0.6857



6. Isosceles cross sections perpendicular to the y -axis.

$$\int_0^2 \frac{1}{2} (e^{y/2} - 1)^2 dy$$

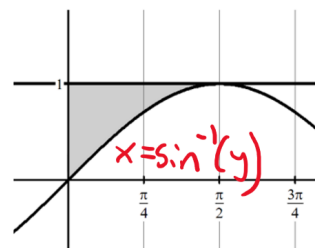
0.7579

7. The y -axis, $y = \sin x$, and $y = 1$ for $0 \leq x \leq \frac{\pi}{2}$. Each cross section perpendicular to the y -axis is a semicircle.

$$\int_0^1 \frac{\pi}{2} \left(\frac{\sin^{-1}(y)}{2} \right)^2 dy$$

$$\frac{\pi}{8} \int_0^1 (\sin^{-1} y)^2 dy \approx$$

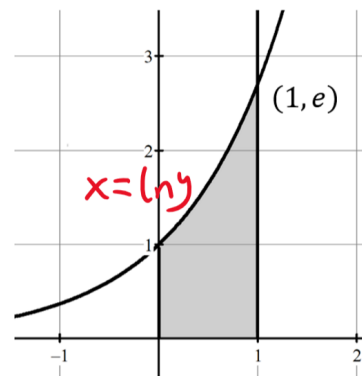
0.1835



8. The region in the first quadrant bounded by $y = e^x$ and the vertical line $x = 1$. The cross sections perpendicular to the y -axis are equilateral triangles.

$$\int_0^1 \frac{\sqrt{3}}{4} (1)^2 dy + \int_1^e \frac{\sqrt{3}}{4} (1 - \ln y)^2 dy$$

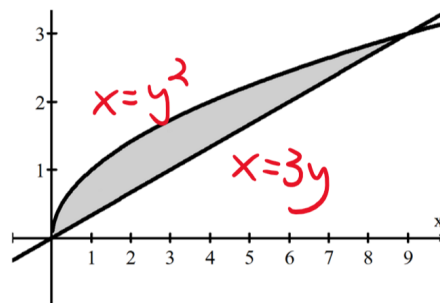
$$0.622$$



9. $y = \sqrt{x}$ and $y = \frac{x}{3}$ cross sections perpendicular to the y -axis are semicircles.

$$\int_0^3 \frac{\pi}{2} \left(\frac{3y - y^2}{2} \right)^2 dy$$

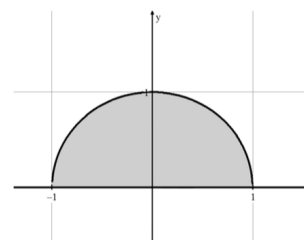
$$\frac{\pi}{8} \int_0^3 (3y - y^2)^2 dy \approx 3.1808$$



10. The x -axis and the graph of $y = \sqrt{1 - x^2}$. Each cross section perpendicular to the x -axis is an isosceles triangle.

$$\int_{-1}^1 \frac{1}{2} (\sqrt{1 - x^2})^2 dx$$

$$\frac{1}{2} \int_{-1}^1 (1 - x^2) dx = \frac{2}{3} \approx 0.6666$$



The following curves create a bounded region. Each solid has cross sections perpendicular to the x -axis that are semicircles. Find the volume of each solid based on the given cross sections. Set up the integral(s) first, then use a calculator to evaluate.

11. $y = x - 4$, $y = 4 - x$, and $x = 0$.

$$x - 4 = 4 - x$$

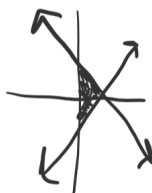
$$2x = 8$$

$$x = 4$$

$$\int_0^4 \frac{\pi}{2} \left(\frac{4 - x - (x - 4)}{2} \right)^2 dx$$

$$\frac{\pi}{8} \int_0^4 (8 - 2x)^2 dx$$

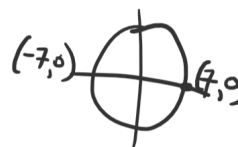
$$33.510$$



12. $x^2 + y^2 = 49$

$$y^2 = 49 - x^2$$

$$y = \pm \sqrt{49 - x^2}$$



$$\int_{-7}^7 \frac{\pi}{2} \left(\frac{\sqrt{49 - x^2} - (-\sqrt{49 - x^2})}{2} \right)^2 dx$$

$$\frac{\pi}{2} \int_{-7}^7 (49 - x^2) dx$$

$$718.3775$$